

Datasheet Servo 6kg 180° Clutch

Manufacturer: DFRobot

Articel-Number: SER0051

Link: [6Kg 180° Clutch Servo with PWM Control \(4.8-6V\) - DFRobot](#)



Figure 1: Servo

INTRODUCTION

DFRobot 6Kg 180° Clutch Servo is a protection-enhanced actuator designed for robotics mechanisms, animatronics, and motion control systems operating under unpredictable loads. Conventional servos often suffer gear damage or overheating when joints are forced externally. This intelligent clutch-enabled servo motor integrates mechanical slip protection and electronic stall safeguards to prevent such failures. Large torque output, stable motion response, and precise PWM angle control make this robotic motion component suitable for educational builds, model systems, and automated prototypes. Direct compatibility with [Arduino](#) servo libraries enables straightforward integration into embedded motion control projects.

MECHANICAL CLUTCH SELF-PROTECTION

This clutch-equipped robotic servo incorporates an internal slip mechanism that activates when external force changes the shaft angle during operation. Rather than transmitting damaging stress to gears or the motor core, the clutch allows temporary disengagement and automatic recovery. After the external disturbance disappears, the actuator repositions itself and resumes commanded motion. Such mechanical protection significantly improves durability in robotic joints, articulated models, and mechanical linkages where unexpected collisions or forced movement frequently occur.

ELECTRONIC STALL PROTECTION

Beyond mechanical protection, this smart servo module integrates electronic overload protection. When the output shaft becomes blocked for approximately five seconds, internal circuitry automatically cuts motor power to prevent overheating and coil burnout. This safety mechanism protects the actuator during heavy mechanical loads, stalled robotic limbs, or jammed linkages. Once the obstruction is cleared and power is reapplied, normal operation resumes, providing a reliable safeguard for experimental robotics and long-running automation tasks.

HIGH TORQUE AND ACCURATE PWM CONTROL

Stable motion output and accurate positioning make this high-torque servo actuator suitable for precise robotic movement. Operating between 4.8V and 6V, the motor delivers stall torque exceeding 6kg·cm at 6V while maintaining smooth response across the full control range. Standard PWM signals with pulse widths from 500–2500µs allow rotation control up to approximately 180°. The absence of an internal limit switch also permits manual continuous rotation handling, supporting mechanical experimentation and flexible integration in servo-driven mechanisms.

EFFICIENT POWER OPERATION AND EASY INTEGRATION

Low operating current and stable electrical characteristics allow this compact servo motor to run from common power sources including battery packs or USB-powered control boards. Straightforward PWM communication enables quick deployment in microcontroller-based systems, robotics prototypes, and educational engineering projects. Large torque output combined with stable performance and simplified installation makes this motion control component well suited for robotic arms, mechanical creatures, interactive models, and embedded automation mechanisms.

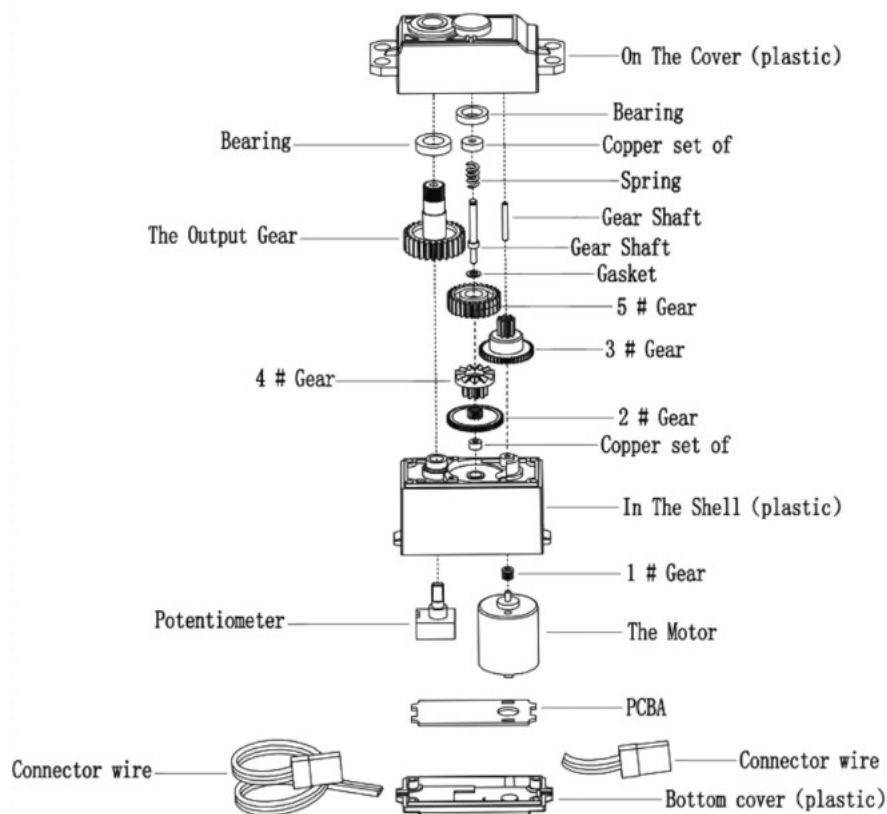


Figure 2: Servo internal structure

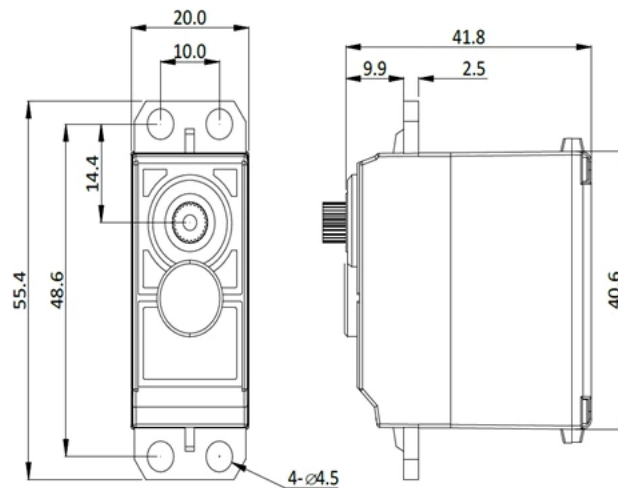


Figure 3: mechanical dimensions

This clutch-protected servo actuator is widely used in robotics joints, mechanical animation systems, intelligent models, and interactive engineering demonstrations. Reliable torque delivery combined with self-protection mechanisms makes this motion control unit suitable for robotics research, maker projects, and STEM Education prototypes requiring durable servo movement.

FEATURES

- Integrated mechanical clutch protection prevents gear damage from external force
- Automatic electronic shutdown after 5 seconds of stall to avoid motor burnout
- Stall torque $\geq 6\text{kg}\cdot\text{cm}$ @6V for strong robotic joint actuation
- Precise PWM control with 500–2500 μs pulse width range
- Operatable rotation angle: $180^\circ \pm 10^\circ$
- Low power consumption suitable for battery or USB power supply
- Compatible with Arduino servo control libraries
- Compact structure for robotics and mechanical model integration

APPLICATIONS

- Robotic arm joints and articulated mechanisms
- Educational robotics and STEM Education platforms
- Animatronics and mechanical creature models
- Interactive exhibition devices
- DIY robotics projects and maker builds
- Model engineering and mechanical linkage experiments

SPECIFICATION

- Operating Voltage: 4.8-6V DC
- Static Current: $\leq 10\text{mA}$ @ 6.0V
- Current(at no load): $\leq 60\text{mA}$ @6.0V
- Current (at stopped): $\leq 1.65\text{A}$ @6.0V
- Rated Torque: $\geq 4.4\text{kg}\cdot\text{cm}$ @6.0V
- Stall Torque: $\geq 6\text{kg}\cdot\text{cm}$ @6.0V
- Operatable Travel: $180^\circ \pm 10^\circ$
- Pulse Width Range: 500~2500 μs
- Communication Mode: PWM